

# Xiaojun (Matt) Cao

Professor

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## I. Education

**Ph.D.** 2004, The State University of New York at Buffalo, Buffalo, NY, USA

Computer Science

Thesis: *Waveband Switching in Wavelength Division Multiplexing Networks*

Advisor: Chunming Qiao

**M.S.** 1999, Institute of Acoustics, Chinese Academy of Sciences, Beijing, China

Electrical Engineering

Thesis: *Optimizing Data Transmission in the DVB System*

Advisor: Ziqiang Hou

**B.S.** 1996, Tsinghua University, Beijing, China

Engineering Physics

## II. Professional Credentials

- **Professor**, 08/2016 – Present, Department of Computer Science, Georgia State University, Atlanta, GA
- **Director of Graduate Studies**, 08/2023 – Present, Department of Computer Science, Georgia State University, Atlanta, GA
- **Director of Graduate Studies**, 08/2017 – 08/2019, Department of Computer Science, Georgia State University, Atlanta, GA
- **Acting Director of Graduate Studies**, 12/2013 – 08/2017, Department of Computer Science, Georgia State University, Atlanta, GA
- **Associate Professor**, 08/2010 – 07/2016, Department of Computer Science, Georgia State University, Atlanta, GA
- **Assistant Professor**, 08/2007 – 07/2010, Department of Computer Science, Georgia State University, Atlanta, GA
- **Assistant Professor**, 07/2004 – 08/2007, College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY

- **Research Assistant**, 08/2001 – 07/2004, Department of Computer Science & Engineering, The State University of New York at Buffalo, Buffalo, NY
- **Software Engineer**, 05/2001 – 08/2001, Brilliant Optical Networks, LLC, Bridgewater, NJ
- **Teaching Assistant**, 09/1999 – 05/2001, Department of Computer Science & Engineering, The State University of New York at Buffalo, Buffalo, NY
- **Software Engineer**, 05/2000 – 08/2000, Cymfony, Inc., Williamsville, NY
- **Software Engineer**, 07/1995 – 05/1997, Beijing Electronic Institute of Asia-Pacific, Beijing, China
- **Research Assistant**, 09/1996 – 09/1999, DSP (Digital Signal Processing) Center, Institute of Acoustics, Chinese Academy of Sciences, Beijing, China

### III. Scholarship and Professional Development

#### A. Grants and Funding

1. “Improvisation Large AI Model,” **Faculty Internal Grants Program: Cycle 1**, \$20,082.00, 2025-2026, **PI**
2. “Enterprise AI and Clinical Analytics,” **Shepherd Center**, \$299,000.00, 2025 – 2030, **PI**, with Dr. Yusen Xia (Co-PI)
3. “DoD Center of Excellence in Advanced Computing and Software (COE-ACS),” DoD Research and Education Program, Contract #W911NF2320224, \$10M, 2023 – 2028, Co-PI, with Dr. Anu Bourgeois (PI), Raj Sunderraman (Co-PI) and other Co-PIs
4. “Investigating BigData Applications in Cyber-Physical Systems (CPS),” **UESTC**, \$45,000.00, 09/2015 – 8/2022, **PI**
5. “Advancements and Universal Usability for Epi-Info Product Suite,” **Centers for Disease Control and Prevention (CDC)**, \$264,816.00, 09/2016 – 12/2017, **Co-PI**, with Dr. Anu Bourgeois (PI) and Raj Sunderraman (Co-PI)
6. “Design and Analysis of Spectrum-sliced Elastic Optical Path Networks,” **National Science Foundation (NSF)** (CNS-1117229), \$197,723.00, 09/2011 – 12/2015, **PI**
7. “Teaching Programming Classes in a Blended Model”, **Center for Instructional Innovation**, **GSU**, \$30,467.00, 07/2013 – 07/2014, **PI**
8. ONTC meeting at OFC’14, **Future Wei Enterprise Co.**, \$1939.04, 2014, **PI**
9. International Workshop on Optical Networking (iWON 2013), **GSU Conference Grants**, \$1900.00, 2013, **PI**
10. International Workshop on Optical Networking (iWON 2013), **Future Wei Enterprise Co.**, \$1000.00, 2013, **PI**
11. “REU: Multi-granular Switching in Optical Networks,” **National Science Foundation (NSF)** (CNV-007805), \$7,810.00, 12/2009 – 11/2010, **PI**
12. GSU Faculty Development Grant, \$800.00, 12/2009
13. “CAREER: A Design of Multi-granular Switching Framework for Optical Networks,” **National Science Foundation (NSF)** (CNS-0813555), \$400,000.00, 2006-2011, **PI**
14. “Advanced Networking Infrastructure Projects for Computing and Collaboration,” **Cisco Research**, \$78,000.00, 2008 - 2009, **PI** for GSU with Dr. G. Khanna of RIT (GSU’s Share is \$15,000.00)

15. "Optimizing Resource Utilization in Wireless Optical Mesh Networks," **GSU RIG**, \$9,884.00, 2008-2009, **PI**
16. GSU Faculty Development Grant, \$800.00, 12/2008
17. GSU Research Travel Grant supporting a trip to the National Science Foundation (NSF) to facilitate the acquisition of extramural funds for a new research project from NETs, \$750.00, 10/2008
18. "Reliable and Prioritized Wireless Sensor & Actor Networks for Industrial Control," **IBM Shared University Research (SUR)** program, \$75,000.00, 2006-2007, **Co-PI**, with Dr. Nirmala Shenoy of RIT as PI
19. "Topology Design and Optimization of High-speed Optical Wireless Communication Networks," RIT FEAD program, \$4,800.00 2005-2006, **PI**
20. "Multi-disciplinary Training on Wireless Ad Hoc & Sensor Networks via Integrated Hardware/Software Experiments," RIT PLIG program, \$12,711.00, 2005-2006, **PI**, with Dr. Nirmala Shenoy and Dr. Fei Hu as Co-PIs

### ***B. Referred Journal Articles***

(Notes: (a). Except for the papers listed below that are marked with \*, the first author is either one of my students at the time or myself. (b). All the magazine papers listed below are peer-reviewed scholarly papers.)

1. D. Zheng, K. Huang, H. Xing, and **X. Cao**, "Towards Latency Differentials Optimization in Deploying URLLC Service Function Chains," *IEEE Transactions on Services Computing*, Feb. 2026, DOI: [10.1109/TSC.2026.3662645](https://doi.org/10.1109/TSC.2026.3662645)
2. C. Peng, Y. Zhao, D. Liu, Z. He, Y. Zhou, and **X. Cao**, "Security-aware Off-site Distributed Pairwise Protection for MoE-based Networks," *IEEE Internet of Things Journal*, Dec. 2025, DOI: [10.1109/JIOT.2025.3611979](https://doi.org/10.1109/JIOT.2025.3611979)
3. D. Zheng, C. Wang, H. Xu, W. Tang, Y. Zhong, and **X. Cao**, "A Provably Efficient In-Network Computing Services Deployment Approach for Security Burst," *Elsevier Computer Networks*, Nov. 2025, DOI: [10.1016/j.comnet.2025.111737](https://doi.org/10.1016/j.comnet.2025.111737)
4. Y. Li, D. Zheng, H. Fang, H. Xing, X. Chen, and **X. Cao**, "Towards Prompt Chain Deployment in Zero Trust-enabled Compute First Networks," *IEEE Transactions on Consumer Electronics*, Aug. 2025, DOI: [10.1109/TCE.2025.3582399](https://doi.org/10.1109/TCE.2025.3582399)
5. C. Wang, D. Zheng, H. Xing, W. Tang, H. Xu, Y. Zhong, and **X. Cao**, "NFV-based Security Estimation and Classification Approaches for AIGC-enabled Edge Networks," *IEEE Transactions on Network Science and Engineering*, Jul. 2025, DOI: [10.1109/TNSE.2025.3593234](https://doi.org/10.1109/TNSE.2025.3593234)
6. D. Zheng and **X. Cao**, "Provably Efficient Service Function Chain Embedding and Protection in Edge Networks," *IEEE/ACM Transactions on Networking*, Vol. 33, No. 1, pp.178-193, Feb. 2025, DOI: [10.1109/TNET.2024.3475248](https://doi.org/10.1109/TNET.2024.3475248)
7. C. Wang, D. Zheng, X. Liu, W. Tang, H. Xu, and **X. Cao**, "Towards Cost Optimization in Security-Aware Service Function Chaining and Embedding over Multi-Vendor Edge Networks," *Elsevier Computer Networks*, Feb. 2025, DOI: [10.1016/j.comnet.2024.111002](https://doi.org/10.1016/j.comnet.2024.111002)

8. J. Yu, H. Yao, K. Ouyang, **X. Cao**, and L. Zhang, "BPS-FL: Blockchain-based Privacy-preserving and Secure Federated Learning," *Big Data Mining and Analytics*, Vol. 8, No.1, pp. 189-213, Feb. 2025, DOI: [10.26599/BDMA.2024.9020053](https://doi.org/10.26599/BDMA.2024.9020053)
9. X. Liu, D. Zheng, H. Xing, L. Feng, C. Peng, and **X. Cao**, "A cost-provable solution for reliable in-network computing-enabled services deployment," *Elsevier Computer Networks*, Feb. 2025, DOI: [10.1016/j.comnet.2024.110997](https://doi.org/10.1016/j.comnet.2024.110997)
10. D. Zheng, H. Xiang, L. Feng, and **X. Cao**, "Provably efficient security-aware service function tree composing and embedding in multi-vendor networks," *Elsevier Computer Networks*, Vol. 254, Dec. 2024, DOI: [10.1016/j.comnet.2024.110843](https://doi.org/10.1016/j.comnet.2024.110843)
11. D. Zheng, H. Fang, S. Cao, Y. Zhong, and **X. Cao**, "Towards Resources Optimization in Deploying Service Function Chains with Shared Protection," *Elsevier Computer Networks*, Vol. 248, Jun. 2024, DOI: [10.1016/j.comnet.2024.110494](https://doi.org/10.1016/j.comnet.2024.110494)
12. C. Peng, D. Zheng, Y. Zhong, and **X. Cao**, "Off-site Protection against Service Function Forwarder Failures in NFV," *Elsevier Computer Networks*, Vol. 221, Feb. 2023, DOI: [10.1016/j.comnet.2022.109510](https://doi.org/10.1016/j.comnet.2022.109510)
13. D. Zheng, G. Shen, Y. Li, **X. Cao**, and B. Mukherjee, "Service Function Chaining and Embedding with Heterogeneous Faults Tolerance in Edge Networks," *IEEE Transaction on Network and Service Management*, Vol. 20, No. 3, pp. 2157-2171, Sep. 2023, DOI: [10.1109/TNSM.2022.3220667](https://doi.org/10.1109/TNSM.2022.3220667)
14. D. Zheng, G. Shen, **X. Cao**, and B. Mukherjee, "Towards Optimal Parallelism-Aware Service Chaining and Embedding," *IEEE Transaction on Network and Service Management*, Vol. 19, No. 3, pp.2063-2077, Sep. 2022, DOI: [10.1109/TNSM.2022.3142184](https://doi.org/10.1109/TNSM.2022.3142184)
15. X. Liao, D. Zheng, Y. Wu, and **X. Cao**, "Coronavirus Pandemic Analysis through Tripartite Graph Clustering in Online Social Networks," *Big Data Mining and Analytics*, Vol. 4, No. 4, pp. 242-251, Dec. 2021, DOI: [10.26599/BDMA.2021.9020010](https://doi.org/10.26599/BDMA.2021.9020010)
16. C. Peng, D. Zheng, S. Philip, and **X. Cao**, "Latency-bounded Off-site Virtual Node Protection in NFV," *IEEE Transaction on Network and Service Management*, Vol. 18, No. 3, pp. 2545-2556, Sep. 2021, DOI: [10.1109/TNSM.2021.3096477](https://doi.org/10.1109/TNSM.2021.3096477)
17. J. Yu\*, Z. Qiao, W. Tang, D. Wang, and **X. Cao**, "Blockchain-Based Decision Tree Classification in Distributed Networks," *Intelligent Automation & Soft Computing (IASC)*, Vol 29, No.3, pp.713-728, Jul. 2021, DOI: [10.32604/iasc.2021.017154](https://doi.org/10.32604/iasc.2021.017154)
18. D. Zheng, H. Gu, W. Wei, C. Peng, and **X. Cao**, "Network Service Chaining and Embedding with Provable Boundaries," in *IEEE Internet of Things Journal*, Vol. 8, No. 9, May 2021, pp.7140-7151, DOI: [10.1109/JIOT.2020.3038617](https://doi.org/10.1109/JIOT.2020.3038617)
19. D. Zheng, C. Peng, X. Liao, and **X. Cao**, "Toward Optimal Hybrid Service Function Chain Embedding in Multiaccess Edge Computing," in *IEEE Internet of Things Journal*, Vol. 7, No. 7, pp. 6035-6045, July 2020, DOI: [10.1109/JIOT.2019.2957961](https://doi.org/10.1109/JIOT.2019.2957961)
20. C. Kong, G. Luo, L. Tian, and **X. Cao**, "Disseminating Authorized Content Via Data Analysis In Opportunistic Social Networks," *Big Data Mining and Analytics*, Vol 2, No.1, pp. 12-24, Mar. 2019, DOI: [10.26599/BDMA.2018.9020028](https://doi.org/10.26599/BDMA.2018.9020028) (Yearly Excellent Paper Award, 2020)
21. B. Camp, J. Mandivarapu, J. Mehta, N. Ramamurthy, J. Wingo, A. G. Bourgeois, **X. Cao**, and R. Sunderraman, "A New Cross-Platform Architecture for Epi-Info Software Suite," *BMC Bioinformatics*, pp. 359:1-8, 2018

22. J. Duan, R. Tian, Y. Xing, Y. Huang, **X. Cao**, Y. Zhang, and G. Zhao, "A Collaborative Pricing Framework for In-Network Caching in Information-Centric Networking", IEEE Access, Vol 6, No.1, pp. 4085-4093, Dec. 2018
23. M. Jalalitabar, E. Guler, G. Luo, L. Tian, and **X. Cao**, "Dependence-Aware Service Function Chain Design and Mapping", IEEE/OSA Journal of Optical Communications and Networking (JOCN), Vol 10, No. 8, Aug. 2018 (invited)
24. F. Hu\*, Q. Hao, Q. Sun, **X. Cao**, R. Ma, T. Zhang, Y. Patil, and J. Lu, "Cyber-Physical System with Virtual Reality for Intelligent Motion Recognition and Training", IEEE Transactions on System, Man, and Cybernetics, Vol. 47, No. 2, pp. 347- 363, Feb. 2017
25. Z. Ye\*, **X. Cao**, J. Wang, H. Yu, and C. Qiao, "Joint Topology Design and Mapping of Service Function Chains for Efficient, Scalable and Reliable Network Function Virtualization," IEEE Network, Vol. 30, No. 3, pp. 81-87, May 2016
26. Yang Wang, Q. Hu, and **X. Cao**, "A Branch-and-Price Framework for Optimal Virtual Network Embedding," Elsevier Computer Networks, Vol. 13, No. 39, Nov. 2015
27. X. Gao\*, Z. Ye, J. Fan, W. Zhong, Y. Zhao, **X. Cao**, H. Yu, and C. Qiao, "Virtual Network Mapping for Reliable Multicast Services with Max-Min Fairness of Reliability," IEEE/OSA Journal of Optical Communications and Networking (JOCN), Vol. 7, No. 9, pp. 942-951, Sep. 2015
28. S. Shakya, **X. Cao**, Z. Ye, and C. Qiao, "Spectrum Allocation in Spectrum-sliced Elastic Optical Path Networks using Traffic Prediction", Springer Photonic Network Communication Journal, Vol. 30, No. 1, pp. 131 – 142, Mar. 2015
29. K. Manandhar, **X. Cao**, F. Hu, and Y. Liu, "Detection of Faults and Attacks Including False Data Injection Attack in Smart Grid Using Kalman Filter," IEEE Transactions on Control of Network Systems, Vol. 1, No. 4, pp. 370 – 379, Dec. 2014
30. Z. Ye\*, X. Li, A. Patel, P. Ji, **X. Cao**, and C. Qiao, "Upgrade-aware Virtual Infrastructure Mapping in Software-Defined Elastic Optical Networks," Springer Photonic Network Communication Journal, Vol. 28, No. 1, pp. 34 - 44, Aug. 2014
31. Y. Zhang, L. Sun, H. Song, and **X. Cao**, "Ubiquitous WSN for Healthcare: Recent Advances and Future Prospects," IEEE Internet of Things Journal, Vol. 1, No. 4, pp. 311-318, Aug. 2014
32. C. Wang, C. Lan, S. Niu, **X. Cao**, and M. Gong, "An ID-Based Certified E-mail Protocol with STTP Suitable for Wireless Mobile Environments," Journal of Computers, Vol. 8, No. 1, pp. 3 - 9, 2013
33. Q. Hu, Yang Wang, and **X. Cao**, "Survivable Network Virtualization for a Single Facility Node Failure: A Network Flow Perspective," Elsevier *Optical Switching and Networking* (OSN), Vol. 10, No. 4, pp. 406-415, 2013
34. Yang Wang, **X. Cao**, A. Caciula and Q. Hu, "Batch Scheduling in Optical Networks," IEEE/OSA Journal of Optical Communications and Networking (JOCN), Vol. 5, No. 2, pp. 116-126, 2013
35. Yang Wang, **X. Cao**, Q. Hu, and Y. Pan, "Towards Elastic and Fine-granular Bandwidth Allocation in Spectrum-sliced Optical Networks," IEEE/OSA Journal of Optical Communications and Networking (JOCN), Vol. 4, No. 11, pp. 906-917, 2012
36. Yang Wang and **X. Cao**, "Multi-granular Optical Switching: A Classified Overview for the Past and Future," IEEE Communications Surveys & Tutorials, Vol. 14, No. 3, pp. 698-713, Jul. 2012

37. T. E. Calhoun, Jr., **X. Cao**, Y. Li, and R. Beyah, "An 802.11 MAC Layer Covert Channel," *Wireless Communications and Mobile Computing (WCMC)*, Vol. 12, No. 5, Apr. 2012
38. Yang Wang and **X. Cao**, "A Study on the Dynamic Waveband Switching in WDM networks," *IEEE/OSA Journal of Optical Communications and Networking (JOCN)*, Vol. 3, No. 5, pp. 390-398, 2011
39. Yang Wang and **X. Cao**, "Multi-granular Waveband Assignment and Protection in WDM Networks," *IEEE/OSA Journal of Lightwave Technology (JLT)*, Vol. 28, No.13, pp. 2004-2013, Jul. 2010
40. L. Pan\*, **X. Cao**, and H. Wu, "Design and Modeling a Distributed and Fair Access MAC Protocol (DFA) for Multi-hop Wireless Networks," *IEEE Transactions on Wireless Communications*, VOL. 8, NO. 5, pp.2434-2442, May 2009
41. **X. Cao**, J. Joseph, J. Li, and C. Xin, "Serialized Batch Scheduling Algorithm for Optical Burst Switching Networks," *IET Journal of Communications*, VOL.3, NO. 3, pp.353-362, Mar. 2009
42. V. Anand\*, **X. Cao**, S. Sheeshia, C. Xin, and C. Qiao, "Fiber-Optical Communication Networks," *Wiley Encyclopedia of Computer Science and Engineering*, <http://doi.wiley.com/10.1002/9780470050118.ecse284>, Sep. 2008
43. **X. Cao**, V. Anand, and C. Qiao, "Waveband Switching for Dynamic Traffic Demands in Multi-granular Optical Networks," *IEEE/ACM Transactions on Networking*, VOL. 15, NO. 5, pp. 957-968, 2007
44. **X. Cao**, V. Anand, and C. Qiao, "A Framework for Waveband Switching in Multi-Granular Optical Networks - Part II," *OSA Journal of Optical Networking*, Vol. 6, No. 1, pp. 48-62, 2007
45. **X. Cao**, V. Anand, and C. Qiao, "A Framework for Waveband Switching in Multi-Granular Optical Networks - Part I," *OSA Journal of Optical Networking*, Vol. 5, No. 12, pp. 1043-1055, Dec. 2006
46. C. Xin\*, B. Wang, **X. Cao** and J. Li, "Logical Topology Design for Dynamic Traffic Grooming in Mesh WDM Optical Networks," *IEEE/OSA Journal of Lightwave Technology (JLT)*, Vol. 24, No.6, pp. 2267- 2275, Jun. 2006
47. F. Hu\*, W. Siddiqui and **X. Cao**, "SPECTRA: Secure Power-Efficient Clustered-Topology Routing Algorithm in Large-scale Wireless micro-Sensor Networks," *International Journal of Information Technology*, Vol. 11, No.2, pp. 95-118, 2005
48. **X. Cao**, V. Anand, J. Li, and C. Xin, "Waveband Switching Networks with Limited Wavelength Conversion," *IEEE Communications Letters*, Vol. 9, No.7, pp.646-648, Jul. 2005
49. X. Yu\*, J. Li, **X. Cao**, Y. Chen, and C. Qiao "Traffic Statistics and Performance Evaluation in Optical Burst Switched Networks," *IEEE/OSA Journal of Lightwave Technology (JLT)*, Vol. 22, No.12, pp. 2722- 2738, Dec. 2004
50. **X. Cao**, V. Anand, Y. Xiong, and C. Qiao, "A Study of Waveband Switching with Multi-layer, Multi-granular Optical Cross-connects," *IEEE Journal on Selected Areas in Communications (JSAC)*, Vol. 21, No. 7, pp.1081-1095, Sep. 2003
51. **X. Cao**, V. Anand, and C. Qiao, "A Waveband Switching Architecture and Algorithm for Dynamic Traffic," *IEEE Communications Letters*, Vol. 7, No. 8, pp. 397-399, Aug. 2003
52. **X. Cao**, V. Anand, and C. Qiao, "Waveband Switching in Optical Networks," *IEEE Communications Magazine*, Vol. 41, No. 4, pp. 105-112, Apr. 2003

### ***C. Book***

53. F. Hu\* and **X. Cao**, “Wireless Sensor Networks: Principles and Practice (Hardcover),” Auerbach Publications; 1<sup>st</sup> edition, 531pages ISBN-10: 1420092154, ISBN-13: 978-1420092158, May 2010

### ***D. Book Chapters***

54. Y. Wang, V. Anand, and **X. Cao**, “Waveband Switching: A Scalable and Cost Efficient Solution for the Internet Backbone,” in “Solutions for Sustaining Scalability in Internet Growth,” IGI Global, 2014, pp. 195-217, doi:10.4018/978-1-4666-4305-5
55. F. Hu\*, Y. Wang, **X. Cao**, Q. Hao, and D. Brown, “Using Wiimote and Kinect for Cognitive Rehabilitation: Toward an intelligent Sensor/Image Signal Processing,” in Book “Cyber-Physical Systems: Integrated Computing and Engineering Design”, Chapter 21, CRC Press, Boca Raton, FL, 2013
56. F. Hu\*, **X. Cao**, D. Brown, J. Park, M. Sun and Y. Wu, “Tele-Rehabilitation Computing: From An Cyber-Physical Perspective,” in Book “Tele-Healthcare Computing and Engineering: Principles and Design,” Chapter 4, CRC Press, Boca Raton, FL, 2013
57. K. Manandhar, **X. Cao**, and F. Hu, “A Framework for Detecting Attacks on Sensors of Water System,” in “Intelligent Sensor Networks: Across Sensing, Signal Processing, and Machine Learning,” F. Hu and Q. Hao Eds, CRC Press, 2012, ISBN: 978-1-4398-9281-7
58. F. Hu\*, **X. Cao**, K. Wilhelm, M. Łukowiak and S. Radziszowski, “NTRU-Based Confidential Data Transmission in Telemedicine Sensor Networks,” in “Security in Ad-hoc and Sensor Networks,” R. Beyah, J. McNair, and C. Corbett Eds, World Scientific Publishing Co., 2009, ISBN: 978-981-4271-08-0
59. **X. Cao**, V. Anand, Y. Xiong, and C. Qiao, “Waveband Switching: A New Frontier in Optical WDM Networks,” in “Optical WDM Networks II: Past Lessons and Path Ahead,” K. Sivalingam and S. Subramaniam Eds, Kluwer Academic Publishers, 2004/2005

### ***E. Refereed Conference Papers***

60. Z. Ding, Y. Wang, Y. Tang, Z. Wang, X. Cao, and C. Peng, “k-Connected Slice Protection for Heterogeneous Concurrent Attacks on AIGC Services,” accepted by IEEE International Conference on Communications (ICC), May. 2026
61. Y. Wu, D. Zheng, J. Ren, H. Xing, W. Tang, and **X. Cao**, “A Runtime- and Cost-Efficient Approach of Deploying Mixture-of-Experts in Edge Networks,” in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2025
62. W. Zeng, D. Zheng, H. Xing, W. Wei, C. Wang, and **X. Cao**, “Cost-Efficient Knowledge Distillation-enabled Student Models Placement in Edge Networks,” in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2025
63. C. Wang, D. Zheng, Y. Zhong, H. Xu, and **X. Cao**, “Heuristic-guided Migration-Agent-Based DRL for Compressed Model Placement in Edge Networks,” in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2025

64. H. Jin, D. Zheng, H. Xing, and **X. Cao**, "Asymmetric Protection-based S-SFC Deployment for LLM-Inference-enabled Networks," in Proc. of IEEE 11th International Conference on Computer and Communications (ICCC), Dec. 2025
65. Y. Zhang, D. Zheng, H. Xing, H. Xu, C. Peng, C. Wang, and **X. Cao**, "Towards Profits Optimization in LLM Inference Model Deployment at the Network Edge," in Proc. of IEEE International Performance, Computing, and Communications Conference (IPCCC), Austin, Texas, Nov. 2025
66. Y. Wu, D. Zheng, J. Ren, J. Wang, C. Wang, C. Peng, and **X. Cao**, "A Cost-and-Time-Efficient Approach of Deploying Mixture of Expert Models," in Proc. of 8th World Conference on Computing and Communication Technologies (WCCCT), Apr. 2025 (Best Paper Award)
67. W. Zeng, D. Zheng, H. Xing, X. Chen, J. Ren, Y. Zhang, C. Peng, and **X. Cao**, "A Cost-Optimization Approach of Placing Student Models in Edge Networks," in Proc. of 8th World Conference on Computing and Communication Technologies (WCCCT), Apr. 2025
68. Y. Zhao, G. Duan, D. Zheng, **X. Cao**, and C. Peng, "Distributed Pairwise Protection for Security-Aware Mission Chains in UAV Networks," in Proc. of 8th World Conference on Computing and Communication Technologies (WCCCT), Apr. 2025 (Best Paper Award)
69. J. Ren, C. Wang, Y. Zhong, S. Cao, D. Zheng, and **X. Cao**, "Towards Expert Models Deployment Cost Optimization in Edge Computing Networks," in Proc. of IEEE International Conference on Communications (ICC) , Jun. 2025, DOI: 10.1109/ICC52391.2025.11161045
70. Y. Li, J. Ren, H. Xing, C. Peng, D. Zheng, and **X. Cao**, "Towards Prompt Chain Deployment Cost Optimization in Zero Trust-enabled Networks," in Proc. of IEEE International Conference on Communications (ICC) , Jun. 2025, DOI: 10.1109/ICC52391.2025.11161045
71. C. Peng, Y. Bai, S. Yang, D. Zheng, and **X. Cao**, "Chain Segment Protection for Dependence-aware Service Function Chain in NFV," in Proc. of 7th World Conference on Computing and Communication Technologies (WCCCT), Apr. 2024, DOI: [10.1109/WCCCT60665.2024.10541465](https://doi.org/10.1109/WCCCT60665.2024.10541465)
72. Y. Zhong, D. Zheng, and **X. Cao**, "A DRL Approach with Network Service Deployment Transformer for Reliable SFC Deployment," in Proc. of IEEE International Conference on Communications (ICC), Denver, Jun. 2024, DOI: [10.1109/ICC51166.2024.10622220](https://doi.org/10.1109/ICC51166.2024.10622220)
73. D. Zheng, S. Cao, H. Xu, and **X. Cao**, "Deploying Security-Aware Service Function Chains with Asymmetric Dedicated Protection," in Proc. of IEEE International Conference on Communications (ICC), Denver, Jun. 2024, DOI: [10.1109/ICC51166.2024.10622657](https://doi.org/10.1109/ICC51166.2024.10622657)
74. D. Zheng, X. Liu, W. Tang, W. Tang, and **X. Cao**, "Cost Optimization in Security-Aware Service Function Chain Deployment with Diverse Vendors," in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2023
75. X. Liu, D. Zheng, and **X. Cao**, "Energy-Efficient In-Network Computing at Mobile Edge Networks," in Proc. of IEEE 9th International Conference on Computer and Communications (ICCC), Dec. 2023
76. C. Peng, D. Zheng, B. Wang, and **X. Cao**, "Off-site Service Function Protection for Type-oriented Forwarder Failures," in Proc. of IEEE International Conference on Communications (ICC), Rome, Italy, May 2023



77. D. Zheng, G. Shen, B. Chen, C. Peng, **X. Cao**, and B. Mukherjee, "Embedding Service Function Chains with Dedicated Protection in Edge Networks," in Proc. of IEEE International Conference on Communications (ICC), Rome, Italy, May 2023
78. B. Wang, **X. Cao**, C. Peng, J. Li, and D. Zheng, "Towards Service Function Chaining and Embedding for Multi-Security Guarantee Levels," in Proc. of 8<sup>th</sup> International Conference on Computer and Communications (ICCC), Dec. 2022
79. D. Zheng, C. Peng, B. Wang, and **X. Cao**, "Towards Deterministic Fault-Tolerant Service Function Slicing in Edge Networks," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Jul. 2022 (**Invited**)
80. C. Peng, D. Zheng, and **X. Cao**, "Latency-aware VNF Protection for Network Function Virtualization in Elastic Optical Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2021
81. D. Zheng, C. Peng, and **X. Cao**, "On the Placement of Edge Server for Mobile Edge Computing," in Proc. of 7<sup>th</sup> International Conference on Computer and Communications (ICCC), Dec. 2021
82. C. Peng, D. Zheng, and **X. Cao**, "Minimum Cost Hybrid Node Protection in NFV," in Proc. of 7<sup>th</sup> International Conference on Computer and Communications (ICCC), Dec. 2021
83. D. Zheng, C. Peng, X. Liao, and **X. Cao**, "Parallelism-aware Service Function Chaining and Embedding for 5G Networks," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Jul. 2021
84. X. Liao, D. Zheng, Y. Wu, and **X. Cao**, "Should We Trust Influencers on Social Networks? On Instagram Sponsored Post Analysis," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Jul. 2021
85. D. Zheng, C. Peng, X. Liao, L. Tian, G. Luo, and **X. Cao**, "Towards Latency Optimization in Hybrid Service Function Chain Composition and Embedding," in IEEE International Conference on Computer Communications (INFOCOM), 2020
86. M. Jalalitabar, Y. Wang, and **X. Cao**, "Branch-Awareness in Service Function Graph Design and Embedding," 2019 IEEE 40th Sarnoff Symposium, Newark, NJ, USA, 2019, pp. 1-3, doi: 10.1109/Sarnoff47838.2019.9067825
87. M. Jalalitabar, Y. Wang, and **X. Cao**, "Branching-Aware Service Function Placement and Routing in Network Function Virtualization," 2019 IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN), Dallas, TX, USA, 2019, pp. 1-6, doi: 10.1109/NFV-SDN47374.2019.9039981
88. S. Philip, C. Peng, and **X. Cao**, "Role Based Medium Access Control in Wireless Sensor Networks," in IEEE International Conference on Computer and Communications (ICCC), 2019
89. D. Zheng, C. Peng, X. Liao, G. Luo, L. Tian, and **X. Cao**, "Service Function Chaining and Embedding with Spanning Closed Walk," in IEEE HPSR, Xi'An, May 2019
90. E. Guler, S. Devaraju, G. Luo, L. Tian, and **X. Cao**, "Multicast-Aware Service Function Tree Embedding," in IEEE HPSR, Xi'An, May 2019
91. D. Zheng, C. Peng, E. Guler, G. Luo, L. Tian, and **X. Cao**, "Hybrid Service Chain Deployment in Networks with Unique Function," in IEEE International Conference on Communications (ICC), Shanghai, May 2019
92. X. Liao, Y. Wu, and **X. Cao**, "Second-order CoSimRank for Similarity Measures in Social Networks," in IEEE International Conference on Communications (ICC), Shanghai, May 2019

93. D. Zheng, E. Guler, C. Peng, G. Luo, L. Tian, and **X. Cao**, "Dependence-Aware Service Function Chain Embedding in Optical Networks," in IEEE International Conference on Communications (ICC), Shanghai, May 2019
94. E. Guler, D. Zheng, G. Luo, L. Tian, and **X. Cao**, "Embedding Multicast Services in Optical Networks with Fanout Limitation," in IEEE International Conference on Communications (ICC), Kansas City, May 2018
95. M. Jalalitabar, E. Guler, G. Luo, L. Tian, and **X. Cao**, "Dependence-Aware Service Function Chain Design and Mapping," in Proc. of IEEE Global Communications Conference (GLOBECOM), Singapore, Dec. 2017
96. E. Guler, D. Zheng, G. Luo, L. Tian, and **X. Cao**, "Virtual Multicast Tree Embedding Over Elastic Optical Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Singapore, Dec. 2017
97. B. Camp, J. Mandivarapu, J. Mehta, N. Ramamurthy, J. Wingo, A. G. Bourgeois, **X. Cao**, and R. Sunderraman, "A Cross-Platform System Architecture for Form Design and Data Analytics for Public Health," in Proc. of ACM BCB, Aug. 2017
98. E. Guler, D. Zheng, G. Luo, L. Tian, and **X. Cao**, "Embedding virtual multicast trees in software-defined networks," in Proc. of IEEE International Conference on Communications (ICC), Paris, France, May 2017
99. C. Kong, G. Luo, L. Tian, and **X. Cao**, "Optimizing Social Connections for Efficient Information Acquisition," in Proc. of IEEE Global Communications Conference (GLOBECOM), Washington, D.C., Dec. 2016
100. E. Guler, G. Luo, K. Koneru, and **X. Cao**, "Closeness-Centrality Based Multicast-Aware Virtual Network Embedding," in Proc. of IEEE Global Communications Conference (GLOBECOM), Washington, D.C., Dec. 2016
101. M. Jalalitabar, G. Luo, C. Kong, and **X. Cao**, "Service Function Graph Design and Mapping for NFV with Priority Dependence," in Proc. of IEEE Global Communications Conference (GLOBECOM), Washington, D.C., Dec. 2016
102. Z. Ye, **X. Cao**, C. Qiao, "Joint Topology Design and Mapping of Service Function Chains in Network Function Virtualization," in Proc. of IEEE Global Communications Conference (GLOBECOM), Washington, D.C., Dec. 2016
103. S. Shakya and **X. Cao**, "Transparent Virtual Network Embedding in Elastic Optical Networks," in Proc. of IEEE Sarnoff Symposium, Princeton, NJ, May 2016
104. X. Gao\*, W. Zhong, Z. Ye, Y. Zhao, J. Fan, **X. Cao**, H. Yu, and C. Qiao, "Virtual Network Mapping for Reliable Multicast Services with Max-Min Fairness," in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2015
105. K. Manandhar and **X. Cao**, "Optimizing Load Schedule for Building Energy Management in Smart Grids," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Aug. 2015
106. C. Kong and **X. Cao**, "Disseminating Authorized Content in Interest-centric Opportunistic Social Networks," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Aug. 2015
107. X. Gao\*, Z. Ye, W. Zhong, C. Qiao, **X. Cao**, H. Zhao, H. Yu, and V. Anand, "Multicast Service-oriented Virtual Network Mapping Over Elastic Optical Networks," in Proc. of IEEE International Conference on Communications (ICC), London, UK, Jun. 2015

108. N. Kadu, S. Shakya and **X. Cao**, "Modulation-aware Multipath Routing and Spectrum Allocation in Elastic Optical Networks," in Proc. of IEEE International Conference on Advanced Networks and Telecommunication Systems (ANTS), Dec. 2014 (**Invited**)
109. S. Shakya, N. Pradhan, **X. Cao**, Z. Ye, and C. Qiao, "Virtual Network Embedding and Reconfiguration in Elastic Optical Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Dec. 2014
110. K. Manandhar and **X. Cao**, "Attacks/Faults Detection and Isolation in the Smart Grid using Kalman Filter," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Aug. 2014
111. Q. Hu, Yang Wang, and **X. Cao**, "Virtual Network Embedding: An Optimal Decomposition Approach," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Aug. 2014
112. C. Kong and **X. Cao**, "Semi-Controlled Authorized Information Dissemination in Content-Based Social Networks," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Aug. 2014
113. K. Manandhar, B. Adcock, and **X. Cao**, "Preserving the Anonymity in MobilityFirst Networks," in Proc. of International Conference on Computer Communications and Networks (ICCCN), Aug. 2014
114. S. Shakya, **X. Cao**, Z. Ye, and C. Qiao, "Spectrum Allocation for Time-varying Traffic in Elastic Optical Networks using Traffic Pattern," in Proc. of Optical Fiber Communication Conference and Exposition (OFC), San Francisco, CA, 2014
115. Yang Wang, Q. Hu and **X. Cao**, "Connectivity as a Service: Towards Optical-based Network Virtualization," in Proc. of International Conference on Computing, Networking and Communications (ICNC), Honolulu, Hawaii, Feb. 2014 (**Invited**)
116. L. Tian, Y. Zhou, and **X. Cao**, "A New Rate-Complexity-QP Algorithm (RCQA) for HEVC Intra-Picture Rate Control," in Proc. of International Conference on Computing, Networking and Communications (ICNC), Honolulu, Hawaii, Feb. 2014
117. K. Manandhar, **X. Cao**, F. Hu, and Y. Liu, "Combating False Data Injection Attacks in Smart Grid Using Kalman Filter," in Proc. of International Conference on Computing, Networking and Communications (ICNC), Honolulu, Hawaii, Feb. 2014
118. S. Shakya, Yang Wang, **X. Cao**, Z. Ye, and C. Qiao, "Minimize Sub-carrier Reallocation in Elastic Optical Path Networks using Traffic Prediction," in Proc. of IEEE Global Communications Conference (GLOBECOM), Atlanta, GA, Dec. 2013
119. C. Kong, **X. Cao**, and M. Liu, "Bayesian-based Video Sharing in Mobile Social Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Atlanta, GA, Dec. 2013
120. Q. Hu, Yang Wang, and **X. Cao**, "Towards Survivable Network Virtualization," in Proc. of IEEE International Conference on Communications (ICC), Budapest, Hungary, 2013
121. Z. Ye, **X. Cao**, X. Gao, and C. Qiao, "A Predictive and Incremental Grooming Scheme for Time-varying Traffic in WDM Networks," in Proc. of IEEE International Conference on Computer Communications (INFOCOM) Mini-conference, Turin, Italy, 2013
122. Q. Hu, Yang Wang, and **X. Cao**, "Resolve the Virtual Network Embedding Problem: a Column Generation Approach," in Proc. of IEEE International Conference on Computer Communications (INFOCOM) Mini-conference, Turin, Italy, 2013

123. S. Shakya and **X. Cao**, "Spectral Defragmentation in Elastic Optical Path Networks using Independent Sets," in Proc. of Optical Fiber Communication Conference and Exposition (OFC/NFOEC), Anaheim, CA, 2013 (**Invited**)
124. Yang Wang, **X. Cao**, and C. Qiao, "Minimize Sub-carrier Re-allocation Overhead in SLICE Networks with Dynamic Traffic," in Proc. of Optical Fiber Communication Conference and Exposition (OFC/NFOEC), Anaheim, CA, 2013
125. Yang Wang, **X. Cao**, A. Caciula, and Q. Hu, "Batch Scheduling in Optical Networks with Feedback/Feed-forward Fiber Delay Lines," in Proc. of IEEE Global Communications Conference (GLOBECOM), Anaheim, CA, 2012
126. A. Rea\*, **X. Cao**, A. Gupta and N. Shenoy, "A Secure Cloud Internetwork Model with Economic and Social Incentives (SCIMES)," in 18th Americas Conference on Information Systems (AMCIS), Washington, Jul. 2012
127. K. Manandhar, **X. Cao**, and F. Hu, "Attack Detection in Water Supply Systems using Kalman Filter Estimator," in Proc. of IEEE Sarnoff Symposium, Princeton, NJ, May 2012
128. Q. Hu, Yang Wang, and **X. Cao**, "Location-constrained Survivable Network Virtualization," in Proc. of IEEE Sarnoff Symposium, Princeton, NJ, May 2012
129. Yang Wang and **X. Cao**, "Wavelength Retuning in Optical Waveband Switching Networks," in Proc. of IEEE International Conference on Computer Communications and Networks (ICCCN), Maui, Hawaii, Jul. 2011
130. Yang Wang, **X. Cao**, and Q. Hu, "Routing and Spectrum Allocation in Spectrum-sliced Elastic Optical Path Networks," in Proc. of IEEE International Conference on Communications (ICC), Kyoto, Japan, Jun. 2011
131. Y. Wang, **X. Cao** and Y. Pan, "A Study on the Routing and Spectrum Allocation in spectrum-sliced Elastic Optical Path Networks," in Proc. of 30<sup>th</sup> IEEE International Conference on Computer Communications (INFOCOM), Shanghai, Apr. 2011, pp. 1503-1511
132. Yang Wang and **X. Cao**, "Distributive Waveband Assignment in Multi-granular Optical Networks," in Proc. of 24<sup>th</sup> IEEE International Parallel & Distributed Processing Symposium (IPDPS), Atlanta, GA, Apr. 2010
133. Yang Wang and **X. Cao**, "Adaptive Waveband Switching in WDM Networks," in Proc. of Optical Fiber Communication Conference and Exposition (OFC/NFOEC) 2010
134. C. Xin\*, G. Hsieh, and **X. Cao**, "Routing and Wavelength Assignment for an Agile Optical Network," in Proc. of International Conference on Electronics and Information Engineering (ICEIE), pp. 432-436, 2010
135. Yang Wang and **X. Cao**, "Non-uniform Waveband Switching in Multi-granular Optical Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Honolulu, Hawaii, Dec. 2009
136. **X. Cao**, Y. Wang, and A. Zelikovsky, "Scheduling Bursts using Interval Graphs in Optical Burst Switching Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Honolulu, Hawaii, Dec. 2009
137. C. Xin\* and **X. Cao**, "A Cognitive Radio Network Architecture Without Control Channel," in Proc. of IEEE Global Communications Conference (GLOBECOM), Honolulu, Hawaii, Dec. 2009
138. **X. Cao**, Yang Wang, Y. Wang, and A. Caciula, "Developing a Multifunctional Network Laboratory for Teaching and Research," in Proc. of ACM SIGITE, Fairfax, VA, Oct. 2009

139. **X. Cao**, B. Chakrabarty, Yang Wang, and F. Hu, "An Energy-efficient Disjoint Path Routing Approach for Wireless Ad-hoc Networks," in Proc. of ISCA 22nd Parallel and Distributed Computing and Communication Systems (PDCCS), Louisville, Kentucky, Sep. 2009
140. **X. Cao**, Y. Wang, Yang Wang, and C. Xin, "Tree-based Burst Aggregation in Optical Burst Switching Networks," in Proc. of ISCA 22nd Parallel and Distributed Computing and Communication Systems (PDCCS), Louisville, Kentucky, Sep. 2009
141. Yang Wang and **X. Cao**, "A New Hierarchical Waveband Assignment Algorithm for Multi-granular Optical Networks," in Proc. of IEEE International Conference on Computer Communications and Networks (ICCCN), San Francisco, CA, Aug. 2009
142. C. Wang, X. Yang, C. Lan, and **X. Cao**, "An Efficient Identity-based Certified E-mail Protocol," in Proc. of Fifth International Conference on Intelligent Information Hiding and Multimedia Signal Processing (IIHMSPP), Kyoto, Japan, 2009
143. Yang Wang and **X. Cao**, "Band-segment Protection in Multi-granular Optical Networks," in Proc. of IEEE Sarnoff 2009, pp. 1-5
144. C. Xin\* and **X. Cao**, "An Agile Lightpath Provisioning Paradigm for IP Over WDM Optical Networks," in Proc. of Optical Fiber Communication Conference and Exposition (OFC/NFOEC) 2008
145. **X. Cao** and C. Xin, "Placement of Multi-Granular Optical Cross-Connects in WDM Networks," in Proc. of Optical Fiber Communication Conference and Exposition (OFC/NFOEC) 2008
146. E. Golen, N. Shenoy, and **X. Cao**, "A Low Latency Scheme for Bulk RFID Tag Reading," in Proc. of IEEE Wireless Communications & Networking Conference (WCNC), Las Vegas, Mar. 2008, pp. 1565-1569
147. **X. Cao**, "Optimizing and Modeling Topology Design in High-speed Optical Wireless Communication Networks," in Proc. of IEEE International Conference on Circuits & Systems for Communications (ICCSC), May 2008, pp. 683-687
148. N. Shenoy\*, **X. Cao**, S. Hild, and P. Chou, "A Turn Taking Medium Access Protocol for Wireless Ad Hoc Networks," in Proc. of Military Communications Conference (MILCOM), Orlando, FL, Oct. 2007, pp. 1-7
149. **X. Cao**, J. Joseph, J. Li, C. Xin, "Group Schedule Serialized Traffic in Optical Burst Switching Networks," in Proc. of IEEE International Conference on Broadband Communications, Networks, and Systems (BROADNETS), Raleigh, NC, Sep. 2007
150. N. Shenoy\*, **X. Cao**, Y. Nozaki, S. Hild, and P. Chou, "Neighbor Turn Taking MAC – A Loosely Scheduled Access Protocol for Wireless Networks," in Proc. of IEEE 18th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC) 2007, pp. 1-5
151. **X. Cao**, S. Liu, L. Pan, and H. Wu, "Enhanced Synchronized Medium Access Control Protocol for Wireless Ad Hoc Networks," in Proc. of IEEE International Conference on Computer Communications and Networks (ICCCN), Honolulu, HI, Aug. 2007, pp. 278-283
152. F. Hu\*, C. May, and **X. Cao**, "Data Aggregation in Distributed Sensor Networks: Towards An Adaptive Timing Control," in Proc. of ITNG 2006, pp. 256-261
153. F. Hu\*, **X. Cao** and S. Kumar, "Trustworthiness in Wireless Sensor and Actuator Networks: Towards Low-complexity Reliability and Security," in Proc. of IEEE Global Communications Conference (GLOBECOM), St. Louis, Dec. 2005, pp. 1696-1700

154. C. Xin, J. Li, **X. Cao**, and B. Wang, "Computing Loss Probability for Dynamic Traffic Grooming in Optical Networks with Wavelength Conversion," in Proc. of IEEE Global Communications Conference (GLOBECOM), St. Louis, Dec. 2005, pp. 291-295
155. C. Xin\*, B. Wang, **X. Cao** and J. Li, "A heuristic logical topology design algorithm for multi-hop dynamic traffic grooming in WDM optical networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), St. Louis, Dec. 2005, pp. 2102-2106
156. C. Xin\*, **X. Cao**, J. Li and B. Wang, "A Route Segment Technique for Blocking Analysis of Dynamic Traffic Grooming," in Proc. of 2<sup>nd</sup> IEEE/Create-Net International Workshop on Traffic Grooming, Co-located with IEEE/Create-Net BroadNets, Boston, Oct. 2005
157. C. Xin\*, B. Wang, **X. Cao** and J. Li, "Formulation of Multi-Hop Dynamic Traffic Grooming in WDM Optical Networks," in Proc. of 2<sup>nd</sup> IEEE/Create-Net International Conference on Broadband Networks (BROADNETS), Boston, Oct. 2005, pp. 1203-1208
158. V. Anand\* and **X. Cao**, "Waveband Switching: A Scalable and Cost Efficient Technique for Optical Network Design," in Proc. of International Conference on Internet Computing (ICOMP), 2005, PP. 225-230
159. F. Hu\* and **X. Cao**, "Security in Wireless Actor & Sensor Networks (WASN): Towards A Hierarchical Re-Keying Design," in Proc. of International Conference on Information Technology Coding and Computing (ITCC), Las Vegas, NV, Apr. 2005. pp. 528-533
160. F. Hu\*, **X. Cao**, and C. May, "Optimized Scheduling for Data Aggregation in Wireless Sensor Networks," in Proc. of International Conference on Information Technology Coding and Computing (ITCC), Las Vegas, NV, Apr. 2005. pp. 557-561
161. J. Li\*, **X. Cao** and C. Xin, "Dual Reservation in Optical Burst Switching Networks," in Proc. of IEEE Sarnoff Symposium, Princeton, Apr. 2005, pp. 180-183
162. **X. Cao**, C. Qiao, V. Anand, and J. Li, "Wavelength Assignment in Waveband Switching Networks with Wavelength Conversion," in Proc. of IEEE Global Communications Conference (GLOBECOM), Dallas, TX, Dec. 2004, pp. 1943-1947
163. **X. Cao**, V. Anand, and C. Qiao, "Multi-Layer versus Single-Layer Optical Cross-connect Architectures for Waveband Switching," in Proc. of 23<sup>rd</sup> IEEE International Conference on Computer Communications (INFOCOM), Hong Kong, Apr. 2004, pp. 1830-1840
164. **X. Cao**, V. Anand, Y. Xiong, and C. Qiao, "Performance Evaluation of Wavelength Band Switching in Multi-fiber All-Optical Networks," in Proc. of 22<sup>nd</sup> IEEE International Conference on Computer Communications (INFOCOM), San Francisco, CA, Apr. 2003, pp. 2251 -2261
165. **X. Cao**, Y. Xiong, V. Anand, and C. Qiao, "Wavelength Band Switching in Multi-granular All-Optical Networks," in Proc. of SPIE Optical Networking & Conference (OPTICOMM), Vol. 4874, Boston, MA, Aug. 2002. pp. 198-210
166. **X. Cao**, J. Li, Y. Chen and C. Qiao, "Assembling TCP/IP Packets in Optical Burst Switched Networks," in Proc. of IEEE Global Communications Conference (GLOBECOM), Taiwan, Nov. 2002. pp. 2808 -2812

#### ***F. Workshop/Poster Papers***

167. C. Kong and **X. Cao**, "Disseminating Content in Interest-centric Opportunistic Social Networks," poster in MobiHoc, Hangzhou, China, 2015

168. Y. Wang, **X. Cao**, A. Caciula, and Q. Hu, "On-line Batch Scheduling in Distributed Optical Networks", in Proc. of 14<sup>th</sup> Workshop on Advances on Parallel and Distributed Processing Symposium (APDCM 2012), joint with IEEE IPDPS, May 2012
169. **X. Cao**, "An Integer Linear Programming Approach to Topology Design in Optical Wireless Communication Networks," in Proc. of EHF-AEROCOMM 2008, joint with IEEE GLOBECOM'08
170. Erik F. Golen, Nirmala Shenoy, and **X. Cao** "A Low Latency Protocol for Bulk RFID Tag Reading," in Proc. of IEEE Upstate NY Workshop on Communications, Sensors, and Networking '07, Nov. 2007
171. N. Shenoy\*, J. Schull, and **X. Cao** "A Design for a Secure Sensor Communications Network," in Proc. of Workshop on Research & Development Partnerships in Homeland Security, Boston, MA, Apr. 26, 2005
172. F. Hu\*, S. Kumar, and **X. Cao** "Trustworthiness in Wireless Sensor and Actuator Networks: Towards Low-complexity Reliability and Security," in Proc. of Fourth IEEE Upstate New York Workshop on Sensor Networks, 2005
173. J. Schull\*, N. Shenoy, **X. Cao**, and D. Johnson, "KarmaNets – A Robust Wireless Mesh Sensor and Communications Network," in Proc. of 1<sup>st</sup> IEEE Upstate NY Communications and Networking Workshop, 2004. pp. 34 – 38
174. **X. Cao** and V. Anand, "Routing and Wavelength Assignment in Waveband Switching Networks," in Proc. of 1<sup>st</sup> IEEE Upstate NY Communications and Networking Workshop, 2004. pp. 137 -141
175. J. Li\*, **X. Cao**, and V. Anand, "Burst Scheduling in Optical Burst Switching Networks," 1st IEEE Upstate NY Communications and Networking Workshop, 2004. pp. 29 – 33
176. **X. Cao**, H. Suen, and L. Troell, "Topology Management in Optical Wireless Communication Networks," in Proc. of RIT GCCIS conference, Dec. 2004

### ***G. Patent***

177. **X. Cao**, C. Qiao, V. Anand, and Y. Xiong, "Efficient Optical Network Design using Multi-granular Optical Cross-connects with Wavelength Band Switching," US. Patents # 7162632, licensed to Gilead Science Institute

### ***H. Invited Talks and Professional Presentations***

1. "Towards Expert Models Deployment Cost Optimization in Edge Computing Networks," in Proc. of IEEE International Conference on Communications (ICC), Montreal, Canada, Jun. 2025
2. "Towards Prompt Chain Deployment Cost Optimization in Zero Trust-enabled Networks," in Proc. of IEEE International Conference on Communications (ICC), Montreal, Canada, Jun. 2025
3. "A DRL Approach with Network Service Deployment Transformer for Reliable SFC Deployment," IEEE International Conference on Communications (ICC), Denver, Jun. 2024

4. "Deploying Security-Aware Service Function Chains with Asymmetric Dedicated Protection," IEEE International Conference on Communications (ICC), Denver, Jun. 2024
5. "Off-site Service Function Protection for Type-oriented Forwarder Failures," IEEE International Conference on Communications (ICC), Rome, Italy, May 2023
6. "Embedding Service Function Chains with Dedicated Protection in Edge Networks," IEEE International Conference on Communications (ICC), Rome, Italy, May 2023
7. "Towards Deterministic Fault-Tolerant Service Function Slicing in Edge Networks", IEEE International Conference on Computer Communications and Networks (ICCCN), Jul. 2022 **(Invited)**
8. "Applications and Challenges in Network Virtualization", IEEE Distinguished Lecture, IEEE Peru Chapter, Sep. 2021 **(Invited)**
9. "Applications and Challenges in Network Virtualization", IEEE Distinguished Lecture, IEEE ComSoc's Chapter in Saudi Arabia, Apr. 2021 **(Invited)**
10. "Applications and Challenges in Network Virtualization", IEEE Distinguished Lecture, IEEE Comsoc Baltimore Chapter, Nov. 2020 **(Invited)**
11. "Service Function Chaining and Embedding", Asia Communications and Photonics Conference (ACP 2020), Oct. 24-27, 2020, Beijing, China **(Invited)**
12. "Applications and Challenges in Network Virtualization", IEEE Distinguished Lecture, IEEE Comsoc North Macedonia Chapter, Oct. 2020 **(Invited)**
13. "Introduction to Network Virtualization", Xidian University, School of Telecommunications Engineering, Jun. 2020 **(Invited)**
14. "Network Function Virtualization", School of Telecommunications Engineering, Xidian University, May 2019 **(Invited)**
15. "Techniques in Future Optical/Electrical Transport Networks", School of Communication & Information Engineering, University of Electronic Science and Technology of China, Jun. 2015 **(Invited)**
16. "Introduction to Big Data for Network Computing and Applications," School of Computer Science and Engineering, University of Electronic Science and Technology of China, June 2015 **(Invited)**
17. "Modulation-aware Multipath Routing and Spectrum Allocation in Elastic Optical Networks," IEEE International Conference on Advanced Networks and Telecommunication Systems (ANTS), Dec. 2014 **(Invited)**
18. "Connectivity as a Service: Towards Optical-based Network Virtualization", IEEE International Conference on Computing, Networking and Communications (ICNC), Feb. 2014 **(Invited)**
19. "Spectral Defragmentation in Elastic Optical Path Networks using Independent Sets," in Proc. of Optical Fiber Communication Conference and Exposition (OFC/NFOEC), Anaheim, CA 2013 **(Invited)**
20. "The Agile Optical Network," ACM Georgia State University Student Chapter meeting, Atlanta, GA, USA, Aug. 30, 2007 **(Invited)**
21. "Multi-granular Waveband Switching in Optical Networks," School of Computing and Information Sciences, Florida International University, Mar. 2007 **(Invited)**
22. "Multi-granular Waveband Switching in Optical Networks," Department of Computer Science, University of Massachusetts Boston, Mar. 2007 **(Invited)**



23. “Multi-granular Waveband Switching in Optical Networks,” IEEE Joint Chapter for Communications and Aerospace technical meeting, Rochester, NY, USA, Feb. 7, 2007 (**Invited**)
24. “Integrated Optical and Wireless Technologies for Broadband Access and Metro Networks Optical Networks” (along with S. Dixit of Nokia Research Center, T. Wang of NEC Laboratories America, and C. Qiao of SUNY at Buffalo), IEEE WoWMoM, Niagara Falls, NY, Jun. 2006 (**Invited**)
25. “Getting Started – Research & Proposal Preparation,” RIT Career Proposal Workshop, Rochester, NY, USA, Nov. 2006 (**Invited**)
26. “Ph.D. Study and Research,” 18th Annual UB CSE Graduate Research Conference, Amherst, NY, USA, Feb. 2005 (**Invited**)
27. “Waveband Switching in Optical WDM Networks, University of Louisiana at Lafayette, Lafayette, LA, USA,” Dec. 3, 2004 (**Invited**)

## IV. Teaching Experiences

- 08/2007 – Present, Teaching graduate and undergraduate courses, Department of Computer Science, Georgia State University, Atlanta, GA
- 12/2005 – 08/2007, Teaching graduate and undergraduate courses, Department of Networking, Security, and Systems Administration (NSSA, a new department created/separated from the IT department), College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY
- 07/2004 - 12/2005, Teaching graduate and undergraduate courses, IT Department, College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY
- 09/1999 – 05/2001, Teaching Assistant for graduate and undergraduate courses, Department of Computer Science & Engineering, The State University of New York at Buffalo, Buffalo, NY

## V. Instructional Activities

- 2022, Faculty sponsor for “Ethical Security and Malware Analysis Guild Club”
- 2011, developed a *new* course: “CSC 8250 Advanced Network Architectures and Protocols”
- 2008, designed and implemented the teaching lab (with Mr. Shaochieh Ou)
- 2008, drafted the proposal for the teaching lab with Dr. Beyah, Dr. Harrison, and Dr. Pan
- 2008, proposed a *new* course: “CSC 8250 Advanced Network Architectures and Protocols,” developed brand new materials for the course CSC 8220 (which had not been offered by the department for years); made significant revisions to the course CSC 4220/6220
- 2/2005- 07/2007 developed *new* course “Wireless Ad-Hoc/Sensor Networks,” Revised course “Data Communications and Computer Networks,” Revised course “OS Scripting”

### A. Courses Taught

| Year | Semester | Course | Size | Course Name |
|------|----------|--------|------|-------------|
|------|----------|--------|------|-------------|

|      |        |               |    |                                              |
|------|--------|---------------|----|----------------------------------------------|
| 2025 | Fall   | CSC8220       | 16 | Advanced Computer Networks                   |
|      |        | CSC8930       | 3  | M.S. Project                                 |
| 2025 | Spring | CSC8930       | 3  | M.S. Project                                 |
| 2024 | Fall   | CSC8220       | 31 | Advanced Computer Networks                   |
|      |        | CSC8930       | 1  | M.S. Project                                 |
| 2024 | Spring | CSC8930       | 3  | M.S. Project                                 |
| 2023 | Fall   | CSC8221       | 31 | Optical/Wireless Networks                    |
|      |        | CSC8930       | 3  | M.S. Project                                 |
| 2023 | Summer | CSC8222       | 15 | Advanced Network Security                    |
| 2023 | Spring | CSC8220       | 34 | Advanced Computer Networks                   |
|      |        | CSC8930       | 3  | M.S. Project                                 |
| 2022 | Fall   | CSC 4222/6222 | 94 | Fundamentals of Cybersecurity                |
|      |        | CSC8930       | 4  | M.S. Project                                 |
| 2022 | Summer | CSC 1302      | 35 | Principles of Computer Programming II        |
| 2022 | Spring | CSC 4222/6222 | 54 | Fundamentals of Cybersecurity                |
|      |        | CSC8220       | 30 | Advanced Computer Networks                   |
|      |        | CSC8930       | 4  | M.S. Project                                 |
| 2021 | Fall   | CSC 4222/6222 | 53 | Fundamentals of Cybersecurity                |
|      |        | CSC8930       | 6  | M.S. Project                                 |
| 2021 | Summer | CSC 1302      | 53 | Principles of Computer Programming II        |
| 2021 | Spring | CSC 4220/6220 | 53 | Computer Networks                            |
|      |        | CSC 1302      | 78 | Principles of Computer Programming II        |
| 2020 | Fall   | CSC8220       | 30 | Advanced Computer Networks                   |
| 2020 | Summer | CSC 1302      | 66 | Principles of Computer Programming II        |
| 2020 | Spring | CSC 4220/6220 | 41 | Computer Networks                            |
|      |        | CSC 1302      | 85 | Principles of Computer Programming II        |
| 2019 | Fall   | CSC 8221      | 12 | Optical/Wireless Networks                    |
|      |        | CSC 8930      | 1  | M.S. Project                                 |
|      |        | CSC 9999      | 1  | Dissertation Research                        |
| 2019 | Summer | CSC 1302      | 37 | Principles of Computer Programming II        |
| 2019 | Spring | CSC 8220      | 29 | Advanced Computer Networks                   |
|      |        | CSC 9999      | 1  | Dissertation Research                        |
| 2018 | Fall   | CSC 4222/6222 | 45 | Introduction to Cyber Security               |
|      |        | CSC 8930      | 2  | M.S. Project                                 |
|      |        | CSC 9999      | 3  | Dissertation Research                        |
| 2018 | Summer | CSC 1302      | 24 | Principles of Computer Programming II        |
| 2018 | Spring | CSC 8220      | 17 | Advanced Computer Networks                   |
|      |        | CSC 9999      | 2  | Dissertation Research                        |
| 2017 | Fall   | CSC 8221      | 11 | Optical/Wireless Networks                    |
|      |        | CSC 8930      | 6  | M.S. Project                                 |
|      |        | CSC 9999      | 2  | Dissertation Research                        |
| 2017 | Summer | CSC 1302      | 34 | Principles of Computer Programming II        |
| 2017 | Spring | CSC 8250      | 26 | Advanced Network Architectures and Protocols |
|      |        | CSC 8900/9900 | 23 | Seminar in Computer Science                  |
|      |        | CSC 8930      | 4  | M.S. Project                                 |

|      |        |                |    |                                              |
|------|--------|----------------|----|----------------------------------------------|
|      |        | CSC9999        | 2  | Dissertation Research                        |
| 2016 | Fall   | CSC 8900/9900  | 51 | Seminar in Computer Science                  |
|      |        | CSC 8930       | 2  | M.S. Project                                 |
|      |        | CSC 9999       | 2  | Dissertation Research                        |
| 2016 | Summer | CSC 2310       | 57 | Principles of Computer Programming II        |
| 2016 | Spring | CSC 8220       | 22 | Advanced Computer Networks                   |
|      |        | CSC 8900/9900  | 27 | Seminar in Computer Science                  |
|      |        | CSC 8930       | 2  | M.S. Project                                 |
|      |        | CSC9999        | 2  | Dissertation Research                        |
| 2015 | Fall   | CSC 8221       | 20 | Optical/Wireless Networks                    |
|      |        | CSC 8900/9900  | 40 | Seminar in Computer Science                  |
|      |        | CSC 8930       | 2  | M.S. Project                                 |
| 2015 | Summer | CSC 2310       | 24 | Principles of Computer Programming           |
| 2015 | Spring | CSC 8250       | 29 | Advanced Network Architectures and Protocols |
|      |        | CSC 8900/9900  | 28 | Seminar in Computer Science                  |
|      |        | CSC9999        | 4  | Dissertation Research                        |
| 2014 | Fall   | CSC 2310       | 72 | Principles of Computer Programming           |
|      |        | CSC 8900/9900  | 37 | Seminar in Computer Science                  |
|      |        | CSC 8930       | 2  | M.S. Project                                 |
|      |        | CSC 9999       | 4  | Dissertation Research                        |
| 2014 | Summer | CSC 2310       | 38 | Principles of Computer Programming           |
| 2014 | Spring | CSC 8220       | 28 | Advanced Computer Networks                   |
|      |        | CSC 8900/9900  | 17 | Seminar in Computer Science                  |
|      |        | CSC 8930       | 3  | M.S. Project                                 |
|      |        | CSC9999        | 3  | Dissertation Research                        |
| 2013 | Fall   | CSC 2310       | 53 | Principles of Computer Programming           |
|      |        | CSC 8221       | 26 | Optical/Wireless Networks                    |
|      |        | CSC 9999       | 3  | Dissertation Research                        |
|      |        | CSC 4220       | 1  | HON Computer Networks                        |
| 2013 | Summer | CSC 2310       | 40 | Principles of Computer Programming           |
| 2013 | Spring | CSC 8250       | 10 | Advanced Network Architectures               |
|      |        | CSC 9999       | 3  | Dissertation Research                        |
| 2012 | Fall   | CSC 2310 - 005 | 46 | Principles of Computer Programming           |
|      |        | CSC 2310 - 015 | 42 | Principles of Computer Programming           |
|      |        | CSC 8930       | 1  | M.S. Project                                 |
|      |        | CSC 9999       | 2  | Dissertation Research                        |
| 2012 | Spring | CSC 8220       | 13 | Advanced Computer Networks                   |
|      |        | CSC 8910       | 5  | Computer Science Topics Seminar              |
|      |        | CSC 9999       | 4  | Dissertation Research                        |
| 2011 | Fall   | CSC 8221       | 9  | Optical/Wireless Networks                    |
|      |        | CSC 9999       | 4  | Dissertation Research                        |
| 2011 | Spring | CSC 8250       | 7  | Advanced Network Architectures and Protocols |
|      |        | CSC 8910       | 5  | Computer Science Topics Seminar              |
|      |        | CSC 8999       | 1  | Thesis Research                              |
|      |        | CSC 9999       | 3  | Dissertation Research                        |

|      |        |               |    |                                                  |
|------|--------|---------------|----|--------------------------------------------------|
| 2010 | Fall   | CSC 2310      | 32 | Principles of Computer Programming               |
|      |        | CSC 9999      | 4  | Dissertation Research                            |
| 2010 | Spring | CSC 8220      | 7  | Advanced Computer Networks                       |
|      |        | CSC 8910      | 9  | Computer Science Topics Seminar                  |
|      |        | CSC 8930      | 1  | M.S. Project                                     |
|      |        | CSC 8999      | 1  | Thesis Research                                  |
|      |        | CSC 9999      | 2  | Dissertation Research                            |
|      |        | CSC 4999      | 1  | Directed Readings                                |
| 2009 | Fall   | CSC 4220/6220 | 17 | Computer Networks                                |
|      |        | CSC 8999      | 2  | Thesis Research                                  |
|      |        | CSC 9999      | 1  | Dissertation Research                            |
| 2009 | Spring | CSC 8220      | 8  | Advanced Computer Networks                       |
|      |        | CSC 8910      | 6  | Computer Science Topics Seminar-Network Analysis |
|      |        | CSC 8930      | 1  | M.S. PROJECT                                     |
|      |        | CSC 8999      | 1  | Thesis Research                                  |
|      |        | CSC 9999      | 1  | Dissertation Research                            |
|      |        | CSC 4220/6220 | 15 | Computer Networks                                |
| 2008 | Fall   | CSC 8930      | 2  | M.S. PROJECT                                     |
|      |        | CSC 8950      | 1  | Direct Research in CS                            |
|      |        | CSC 9999      | 1  | Dissertation Research                            |
|      |        | CSC 8220      | 14 | Advanced Computer Networks                       |
| 2008 | Spring | CSC 8910      | 3  | Computer Science Topics Seminar                  |
|      |        | CSC 4220/6220 | 11 | Computer Networks                                |
| 2007 | Fall   | 4050-402      | 29 | OS Scripting                                     |
| 2007 | Spring | 4002-897      | 2  | Thesis Guidance                                  |
|      |        | 4050-402      | 30 | OS Scripting                                     |
| 2006 | Winter | 4050-341      | 31 | Data Communications and Computer Networks        |
| 2006 | Fall   | 4050-582/782  | 11 | Wireless Ad-Hoc/Sensor Networks                  |
|      |        | 4050-599      | 2  | Independent Study                                |
| 2005 | Winter | 4050-582/782  | 16 | Wireless Ad-Hoc/Sensor Networks                  |
|      |        | 4050-342      | 16 | Principles of Networking                         |
| 2005 | Fall   | 4002-341      | 24 | Data Communications and Computer Networks        |
|      |        | 4002-342      | 17 | Principles of Networking                         |
|      |        | 4002-519      | 6  | Network Troubleshooting                          |
|      |        | 4002-522/822  | 5  | Introduction to Network Programming              |
| 2005 | Spring | 4002-421      | 20 | Systems Administration I                         |
|      |        | 4002-342      | 21 | Internetworking Lab                              |
|      |        | 4002-402      | 25 | OS Scripting                                     |
|      |        | 4002-340      | 12 | Computer Concepts and Software Systems           |
| 2004 | Winter | 4002-342      | 14 | Internetworking Lab                              |
|      |        | 4002-341      | 11 | Data Communications and Computer Networks        |
|      |        | 4002-342      | 34 | Internetworking Lab                              |

## ***B. Student Supervision***

### **Ph.D. students supervised**

- (1) Yang Wang (2012), Dissertation: “Resource Management in Survivable Multi-Granular Optical Networks,” Assistant Professor, La Salle University, Philadelphia, PA (was with Internap Corp. upon graduation).
- (2) Qian Hu (2015), Dissertation: “Towards a Virtualized Next Generation Internet.”
- (3) Kebina Manandhar (2015), Dissertation: “Enhancing Security in the Future Cyber Physical Systems,” Department of Computer Science, GSU
- (4) Sunny Shakya (2015), Dissertation: “Management of Spectral Resources in Elastic Optical Networks,” Ciena Corporation, Alpharetta, GA
- (5) Chenguang Kong (2016), Dissertation: “Content Dissemination in Mobile Social Networks,” FICO, San Diego, CA
- (6) Maryamsadat Jalalitar (2018), Dissertation: “Service Function Graph Design and Embedding in Next Generation Internet,” Boston College
- (7) Evrim Guler (2019), Dissertation: “Multicast Aware Virtual Network Embedding In Software Defined Networks”
- (8) Danyang Zheng (2021), Outstanding Graduate Student Award 2021, Dissertation: “Network Function Virtualization Service Delivery for Future Internet”
- (9) Xueting Liao (co-advised with Dr. Yubao Wu) (2021), Dissertation: “Graph Mining and its Applications in Online Social Networks”
- (10) Chengzong Peng (2022), Dissertation: “Service Function Protection in Network Function Virtualization”
- (10) Yihan Zhong
- (11) Chao Wang (GSSP Dean’s Graduate Research Grants Award 2025)
- (12) Jacob Cartwright
- (13) Anubha Mittal

### **Ph.D. thesis committee member**

- 2026: (1) Parsa Ghazvinian
- 2025: (1) Alireza Marefatvayghani, (2) Chunlan Gao
- 2023: (1) Khadija Ashraf, (2) Brendan B. Camp
- 2022: (1) MD Rashed Rahman, (2) Ramya Tekumalla, (3) Jaya Krishna Mandivarapu
- 2021: (1) Christopher Freas, (2) Sudha Tushara Sadasivuni
- 2016: (1) Goutham Kamath; (2) Guoliang Liu; (3) Lei Shi
- 2013: Shouling Ji
- 2009: Vu Trung Chinh

### **M.S. thesis/project committee chair**

- 2025: (1) Prashanth Sai Konda, (2) Chandler Dugan, (3) Nikhitha Sri Todeti
- 2024: (1) Divya Sri Chakravaram, (2) Chandana Movva, (3) Dattu Reddy Maddur
- 2023: Bhargav Muppalla
- 2022: (1) Davidson Fleurantin, (2) Sowjanya Chintala, (3) Manichandan Valgot, (4) Surya Teja Yellutla, (5) Sonal Chavan, (6) Pavan Shivalingaiah
- 2021: (1) Amrita Biswas, (2) Saidarao Chirumamilla, (3) Rohan Kodangal Varala, (4) Krishna Mani Teja Katragadda, (5) Venus Pagidimarri
- 2020: (1) Veera Venkata Subramanya Aditya Bhamidipati, (2) Aishwarya Rajuladevi Manikyam

- 2019: Sheng Li
- 2018: (1) Chien Tung Chen; (2) Shubin Wu; (3) Swaroop Devaraju
- 2017: (1) Dileep Gunda; (2) Swetha Gatty; (3) Akshita Maradapu Vera Venkata Sai; (4) Siddarth Shekhar; (5) Brendan B. Camp; (6) Yung Tseng
- 2016: (1) Megan Smith; (2) Divya Aluvala; (3) Sri Shravya Basavaraju; (4) Kaushik Koneru
- 2015: Chyeeka Brown
- 2014: (1) Nabina Pradhan; (2) Hari Uday; (3) Jyoti Kumari; (4) Neha Kadu; (5) Stephen Mathew; (6) Quintin Ash
- 2012: (1) Barnali Chakrabarty; (2) Yufan Wang
- 2011: Vijetha Shivarudraiah
- 2010: Rahul Mallampati
- 2009: (1) Ranjitha Shivarudraiah; (2) Yichuan Wang
- 2007: Frank Suen
- 2006: Shenbo Liu

#### **M.S. thesis/project committee member**

- 2025: Lakshmi Naga Sai Sravani Kota
- 2024: Weizhen Liu
- 2023: (1) Ishu Goyal; (2) Sai Krishna Reddy Kandhadi
- 2022: (1) Varchaleswari Ganugapati; (2) Harsha Duddu
- 2021: (1) AbdulHaseeb Ahmed; (2) Kiruthiga Sekar; (3) Xinyu Hu; (4) Hsiu Yuan Fan
- 2019: (1) Huaifu Hu; (2) Carlos Enrique Hearn
- 2016: (1) Tri Cao Hoang; (2) Valli Srilakshmi Devalla
- 2015: (1) Paritosh P. Ramanan; (2) Monjur Alam
- 2014: (1) Sri Vidya Yechuri; (2) Syambabu Pothini; (3) Srinivasa Satya Pradeep Nekkalapudi; (4) Kiran Sagar Kuruba; (5) Sree Vidya Naraharasetti; (6) Naga Manojna Chintapalli
- 2013: (1) Pridhvi Raj Ramanujula; (2) Xuhong Zhang; (3) Ben Adcock
- 2012: Syed Haque
- 2009: (1) Marco Valero; (2) Telvis Calhoun
- 2008: (1) Aravind Venkataraman; (2) James Reed Newman
- 2007: Daisuke Oka
- 2006: (1) Meng Jiang; (2) Bhushan Mehendale; (3) Thomas J. Lantier; (4) Justin J. Madigan

#### **Visiting scholars under supervision**

- Yu Chen 2019-2020
- Shufen Niu 2018-2019
- Jie Duan 2018-2019
- Yuanyuan Zhang 2018-2019
- Yu Hu 2016-2016
- Bangju Wang 2014-2015
- Sung-Woon Lee 2014-2015
- Qifei Zhang 2014-2015
- Min Mao 2014-2015
- Huawei Zhang 2013-2014

|               |           |
|---------------|-----------|
| ▪ Yuan Zhang  | 2013-2014 |
| ▪ Ling Tian   | 2013-2014 |
| ▪ Yimin Zhou  | 2013-2013 |
| ▪ Ming Liu    | 2012-2013 |
| ▪ Caifen Wang | 2010-2011 |
| ▪ Chaoyang Li | 2009-2010 |

## VI. Department and College Service

|              |                                                                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-2025    | University Senate Committee<br>University Committee on Admissions & Standards<br>University Committee on Academic Programs                                                                      |
| 2023-Present | College Graduate Council                                                                                                                                                                        |
| 2023-Present | Graduate Committee                                                                                                                                                                              |
| 2022-2023    | University Review Committee for the Stamps and Presidential Scholarships<br>Ad-hoc School of Computing Plan Committee<br>Strategic Planning Committee<br>Academic Professional Search Committee |
| 2021-2023    | University Senate Committee<br>University Committee on Research<br>University Committee on Planning and Development<br>Executive Committee, Chair                                               |
| 2021-2022    | Academic Professional Search Committee, Chair<br>Ad hoc Curriculum Committee                                                                                                                    |
| 2019-2021    | University Senate Committee<br>Committee on Academic Programs<br>CAP Subcommittee: Graduate Council<br>Committee on Planning and Development                                                    |
| 2016-present | Promotion & Tenure Committee                                                                                                                                                                    |
| 2020-2022    | Faculty Search Committee                                                                                                                                                                        |
| 2020-2021    | Bylaws Committee, Chair                                                                                                                                                                         |
| 2019-2020    | Faculty Search Committee                                                                                                                                                                        |
| 2017-2019    | Director of Graduate Studies                                                                                                                                                                    |
| 2014-2019    | College Graduate Council                                                                                                                                                                        |
| 2014-2019    | Graduate Committee, chair                                                                                                                                                                       |
| 2017-2018    | College P&T Area Committee                                                                                                                                                                      |
| 2017-2018    | College Strategic Planning Working Group                                                                                                                                                        |
| 2016-2017    | Faculty Search Committee, Chair                                                                                                                                                                 |
| 2014-2017    | Acting Director of Graduate Studies                                                                                                                                                             |
| 2016-2017    | Academic Program Review Committee                                                                                                                                                               |
| 2014-2015    | Bylaws Committee, Chair                                                                                                                                                                         |
| 2011-2015    | GSU Internal Grants Review Committee                                                                                                                                                            |
| 2011-2012    | Triennial Chair Evaluation Committee                                                                                                                                                            |
| 2010-2014    | Executive Committee                                                                                                                                                                             |

|           |                                             |
|-----------|---------------------------------------------|
| 2010-2011 | Promotion & Tenure Committee                |
| 2007-2014 | Faculty Search Committee                    |
|           | Curriculum Committee                        |
|           | Library Committee                           |
|           | Graduate Committee                          |
|           | Ph.D. Qualifying Examination Committee      |
| 2006-2007 | College Ph.D. Committee                     |
|           | Assessment Committee                        |
| 2005-2007 | Curriculum Committee                        |
| 2005-2007 | Faculty Search Committee                    |
| 2004-2006 | College Ph.D. Program Preparation Committee |
| 2004-2005 | IT Department Facilities Committee          |
|           | ANSA Assessment Committee                   |

## VII. Professional Services/Activities

- Senior Member of IEEE and its Communications Society, ACM
- IEEE ComSoc TC Restructure Sub-Committee (2022-2024)
- IEEE Distinguished Lecture (2020-2022)
- Nomination and Election (N&E) Subcommittee, IEEE Optical Networking Technical Committee (ONTC) (2021)
- Chair of the IEEE Optical Networking Technical Committee (ONTC) (2017-2019)
- Vice Chair of the IEEE Optical Networking Technical Committee (ONTC) (2015-2017)
- Elected Secretary of the IEEE Optical Networking Technical Committee (ONTC) (2013-2015)

### Review Panels

- Canada NSERC Discovery (2018-2019, 2020, 2021)
- NASA Postdoctoral Program Review (2011, 2012)
- National Science Foundation (NSF) (2007, 2013, 2017)
- NSF Global Environment for Network Innovations (GENI) (2008, 2009, 2010, 2013, 2022, 2024-05)
- US Department of Energy (2008, 2009, 2010, 2012, 2015)
- **Proposal Reviewing**, for Rochester, New York WXXI Public Broadcasting Council, 2006

### Editorships

- **Guest Editor**, IEEE Wireless Communications (2022 – 2023)
- **Editor**, Elsevier Computer Networks (since 2019)
- **Associate Editor**, Springer Photonic Network Communications Journal (since 2013)
- **Associate Editor**, Elsevier Computer Communications (since 2013)
- **Associate Technical Editor**, IEEE Communications Magazine (2017 - 2022)
- **Associate Editor**, IEEE Communications Letters 2009-2014
- **Guest Editor**, Tsinghua Science and Technology, 2012



### **Conference/Workshop Organizing Committee**

- ICNC'26, Invited Talk Chair
- IEEE HPSR'20, Technical Program Co-Chair
- ICNC'20 OGC, Symposium Co-Chair
- IEEE ICC'19, ONS Symposium Co-Chair
- IEEE BDCloud 2016, Technical Program Co-Chair
- IEEE GLOBECOM'16 NGN Symposium Co-Chair
- Publicity Chair, PICom 2014
- Local Chair, 2<sup>nd</sup> International IBM Cloud Academy Conference ICA CON 2014
- Workshop Chair, iWON'13, December, Atlanta, GA, USA
- Selection Committee of the 2013 IEEE ComSoc Student Competition "Communications Technology Changing the World"
- Workshop Co-chair, CSA-13, Dec. 18-21, 2013, Danang, Vietnam
- Symposium Co-Chair, International Conference on Computing, Networking and Communications (ICNC) 2013
- Program Co-Chair, The 4<sup>th</sup> FTRA International Conference on Computer Science and its Applications 2012
- Program Co-Chair, International Conference on Algorithms and Architectures for Parallel Processing 2010
- Track Co-Chair, AMCIS 2011
- Organizing Committee, RIT-VoIP workshop 2006, IEEE Upstate NY Workshop on Communications and Networking 2005; RIT VoIP Mini-Summit 2005; 1st and 2nd International Workshop on Optical Burst Switching
- Vice-chair, IEEE Joint Chapter on Communications and Aerospace, Rochester, 2005-2006

### **Session Chair**, for conferences and workshops

- INFOCOM'20, HPSR'20, ICC 2019, ICC 2018, INFOCOM 2017, GLOBECOM 2016, GLOBECOM 2014, ICCCN 2014/2015, ICCCN 2013

### **Technical Program Committee**, for dozens of conferences and workshops, including

- IEEE INFOCOM: 2022, 2023, 2024, 2025, 2026
- **2021**, CRC Book Review
- **2020**: IEEE INFOCOM'21, HPSR'21
- **2019**: IEEE INFOCOM'20, HPSR'20
- **2018**: IEEE ICC'19, GLOBECOM'19
- **2017**: IEEE INFOCOM'18, UB ETC'17, IEEE ICC'18
- **2016**: IEEE INFOCOM'17, TECON'17, IEEE ICC'17
- **2015**: IEEE ICC'16, NaNA2016, IEEE INFOCOM'16
- **2014**: IEEE INFOCOM'15, IEEE INFOCOM'14,
- **2013**: IEEE INFOCOM'13, IEEE GLOBECOM'13
- **2012**: IEEE INFOCOM'12, IEEE ICC'12, IEEE GLOBECOM'12
- **2011**: IEEE ICC'11, IEEE GLOBECOM'11, COIN/PS 2011, IEEE HSN'11, iCOST'11, ISPA'2011
- **2010**: IEEE GLOBECOM'10, IEEE ICCCN'10, LOCALGOS'10, ACP 2010

- **2009:** IEEE ICC'09, ICA3PP 2009, APOC'09
- **2008:** IEEE ICC'08, IEEE GLOBECOM'08, IPDPS'08, LCN'08, ICA3PP'08, PDCAT'08,
- **2007:** IEEE ICC'07, IEEE GLOBECOM'07, IEEE WLN'07, IEEE ICCCN'07, IEEE BROADNETS'07,
- **2006:** IEEE ICC'06, IEEE GLOBECOM'06, BROADNETS'06, CAMPS'06, SensorWare'06, IEE Mobility Conference'06
- **2005:** IEEE GLOBECOM'05, ICETE2005, IEEE WirelessCOM 2005, IWOBBS'05

**Panel Speaker,** Panel on Integrated Optical and Wireless Technologies for Broadband Access and Metro Networks, Optical Networks (along with S. Dixit of Nokia Research Center, T. Wang of NEC Laboratories America, and C. Qiao of SUNY at Buffalo), IEEE WoWMoM, Niagara Falls, NY, Jun. 2006

**Ad Hoc Referee for:** Books from Springer Publisher, John Wiley & Sons, CRC, many journals/magazines and conferences/workshops including: IEEE/ACM Transactions on Networking, IEEE Transactions on Human-Machine Systems, IEEE Transactions on Computers, ACM, Transactions on Privacy and Security, IEEE Internet of Things Journal, IEEE Transactions on Network Science and Engineering, IEEE Journal on Selected Areas in Communications (JSAC), IEEE Transactions on Network and Service Management(TNSM), IEEE Transactions on Parallel and Distributed Systems(TPDS), Journal of High-Speed Networks, ACM/Kluwer Mobile Networks and Applications (MONET), OSA Journal of Optical Communications and Networking (JOCN), Journal of Computer Science and Technology(JCST), Elsevier Journal of Optical Switching and Networking (OSN), Journal of Network and Systems Management(JNSM), IEEE/OSA Journal of Lightwave, Journal of Computer Network, International Journal of Computers and Applications, Optics Express Journal, Journal of Supercomputing, Journal of the Network and Systems Management (JONS), Journal of Network and Computer Applications (JNCA), IEEE Computer Networks, IEEE Transactions on Computers, IEEE Photonics Technology Letters, IEEE Communication Magazine, IEEE Communication Letter, Wiley Wireless Communications and Mobile Computing, Elsevier Computer Networks, IEEE IPDPS08, IEEE Infocom04, IEEE Infocom05, IEEE Globecom05, IEEE Globecom06, IEEE ICC05, IEEE ICC06, IEEE BroadNets05, IEEE BroadNets06, ONDM05, ITCC05, ITRE05, ICETE05, IPSI2005, ADCOM 05, WOBBS 2005, WOBBS2006 ISWPC 2006, IEEE CHINACOM-Comm'2006, MILCOM'10

## VIII. Awards and Honors

- Excellent Paper Award, Big Data Mining and Analytics Journal, 2020
- NSF CAREER Award, 2006–2011
- New York State Professional Development Award, 2003
- IEEE Communications Society Grants for attending INFOCOM, 2003
- Teaching/research assistantship and full tuition scholarship from the CSE department, State University of New York at Buffalo, fall 1999 - spring 2004
- Student Travel Award, CSE Graduate Student Association, SUNY at Buffalo, 2002 & 2003
- SPIE Student Travel Award for attending OptiComm Conference, 2002

- Student Travel Award, UB Graduate Student Association, SUNY at Buffalo, 2002
- Excellent Master's thesis award from DSP Center, Institute of Acoustics, Chinese Academy of Sciences, 1999
- Awarded Excellent Student Scholarship by Tsinghua University for four school years, 1991–1996